

Mini Guide 17: Intangible Assets — Recording, Reporting, and Reading Them as an Analyst

FIN 4000 Investments • BKM 13e Ch. 1, §1.1 (real vs. financial assets), with Ch. 19 (financial statement analysis)

Big picture: intangibles are real assets because of what they do, not what they are made of. Accounting, however, records only some of them. This guide shows what GAAP and IFRS recognize, how the gap between recorded and unrecorded intangibles distorts the ratios you use in Ch. 19, and how to adjust for it.

Learning Objectives

By the end of this guide you should be able to: 1. Explain why intangibles are **real assets** (BKM §1.1) even though accounting calls them “intangible.” 2. State the IFRS (IAS 38) and US GAAP (ASC 350) definitions and apply the **identifiability** test. 3. Distinguish **identifiable intangibles** from **goodwill** and compute each in a purchase price allocation. 4. Compute **tangible book value per share** and compare P/B with P/TBV. 5. Show how **capitalizing vs. expensing** development costs changes net income, book equity, ROE, and P/B. 6. Identify the quality-of-earnings signals tied to goodwill and intangible impairments.

1. Core Concept: Real vs. Financial Assets (BKM Ch. 1, §1.1)

Real assets produce goods, services, or cash flows directly. **Financial assets** are claims on the income generated by real assets and net out to zero for the economy as a whole. A patent, brand, or body of technical know-how earns income through use, so it is a real asset. A factory does the same thing in steel and concrete.

BKM §1.1 makes the same point: a society's wealth rests on its real assets, while financial assets only determine how ownership of that wealth is distributed among investors. BKM's own examples of real assets include the knowledge used to produce goods and services, which is exactly the intangible case.

Two classification schemes run in parallel:

Scheme	Question it asks	Intangibles land in...
Finance: real vs. financial	Does it produce value directly, or is it a claim on something that does?	Real
Accounting: tangible vs. intangible	Does it have physical substance?	Intangible

2. Core Concept: What the Standards Say

IFRS (IAS 38). An intangible asset is (1) controlled by the entity, with expected future economic benefits, (2) without physical substance, and (3) **identifiable**: either *separable* (can be sold, licensed, rented, or exchanged) or arising from *contractual or legal rights*. Goodwill acquired in a business combination fails the identifiability test and falls under IFRS 3 instead.

US GAAP (ASC 350). The FASB Master Glossary defines intangible assets by exclusion: assets, not including financial assets, that lack physical substance. Goodwill is treated separately. GAAP applies an IFRS-like identifiability test at acquisition (ASC 805).

Issue	IFRS (IAS 38)	US GAAP (ASC 350/730)
Research Development	Expensed Capitalized once six criteria are met	Expensed Generally expensed (software is the main exception)
Revaluation	Allowed if an active market exists (rare)	Not allowed
Indefinite-lived intangibles	Not amortized; impairment tested	Not amortized; impairment tested
Goodwill (public companies)	Not amortized; impairment tested	Not amortized; impairment tested

3. Core Concept: The Analyst's View (BKM Ch. 19)

- **Recorded vs. unrecorded.** Internally generated brands, know-how, and customer relationships are expensed as incurred and never reach the balance sheet. The *same* asset bought in an acquisition is capitalized. Book value therefore understates intangible-heavy firms.
- **Identifiable intangibles vs. goodwill.** Identifiable intangibles with finite lives are amortized (a predictable earnings drag). Goodwill is a residual that sits on the balance sheet until impaired.
- **Comparability.** BKM's discussion of comparability problems highlights quality of earnings, fair-value estimates, and international differences in accounting conventions. Intangibles are one of the clearest cross-border examples.
- **ROE decomposition.** BKM breaks ROE into five factors:

$$\text{ROE} = \underbrace{\frac{\text{Net income}}{\text{Pretax profit}}}_{\text{Tax burden}} \times \underbrace{\frac{\text{Pretax profit}}{\text{EBIT}}}_{\text{Interest burden}} \times \underbrace{\frac{\text{EBIT}}{\text{Sales}}}_{\text{Margin}} \times \underbrace{\frac{\text{Sales}}{\text{Assets}}}_{\text{Turnover}} \times \underbrace{\frac{\text{Assets}}{\text{Equity}}}_{\text{Leverage}}$$

Capitalizing development costs changes the margin (less expense now, amortization later), turnover and leverage (a larger asset base), and equity (retained earnings are higher).

Key equations

$$\text{TBVPS} = \frac{\text{Equity} - \text{Goodwill} - \text{Other intangibles}}{\text{Shares outstanding}}$$

$$\text{Goodwill} = \text{Price paid} - \text{FV of net identifiable assets}$$

4. Worked Examples

*BA II Plus note: the calculator ships in **chain (Chn)** mode, which evaluates strictly left to right. Examples 1–3 work in either mode. Example 4 mixes multiplication and addition, so switch to AOS first: press 2nd FORMAT, press ↓ until “Chn” shows, 2nd SET to toggle to “AOS”, 2nd QUIT.*

Example 1 — Tangible book value per share (bank)

A bank reports equity of \$8,400M, goodwill of \$2,100M, other intangibles of \$650M, and 250M shares. The stock trades at \$30. Compute P/B and P/TBV.

- Book value per share: $8,400 \div 250 = \$33.60$, so $P/B = 30 \div 33.60 = 0.89$
- Tangible book value: $(8,400 - 2,100 - 650) \div 250 = \22.60 , so $P/TBV = 30 \div 22.60 = 1.33$

Keystrokes: $8400 - 2100 - 650 = \div 250 = \rightarrow \mathbf{22.60}$; then $30 \div 22.6 = \rightarrow \mathbf{1.33}$.

Reading it: the stock looks cheap on P/B (below 1.0) but not on P/TBV, because roughly a third of book equity is acquisition-related intangibles.

Example 2 — Purchase price allocation and amortization

A firm buys a target for \$400M. Net tangible assets have a fair value of \$230M. Appraisers identify \$50M of customer relationships with a 10-year life. Compute goodwill and the annual after-tax earnings effect (tax rate 25%; ignore deferred taxes).

- Goodwill = $400 - 230 - 50 = \$120\text{M}$ (residual; not amortized, impairment tested)
- Annual amortization = $50 \div 10 = \$5\text{M}$
- After-tax earnings drag = $5 \times (1 - 0.25) = \$3.75\text{M}$ per year

Keystrokes: $400 - 230 - 50 = \rightarrow \mathbf{120}$; $50 \div 10 = \rightarrow \mathbf{5}$; $\times .75 = \rightarrow \mathbf{3.75}$.

Example 3 — Capitalize vs. expense development costs

Two otherwise identical firms start the year with equity of \$1,000M and earn \$300M before development costs. Each spends \$100M on development. Firm A (US GAAP) expenses it. Firm B (IFRS) capitalizes it and amortizes straight-line over 5 years. Tax rate 25%; ignore deferred taxes and dividends.

	Firm A (expense)	Firm B (capitalize)
Development expense in year 1	\$100M	\$20M
Pretax income	\$200M	\$280M
Net income	\$150M	\$210M
Capitalized asset at year-end	\$0	\$80M
ROE (on beginning equity)	15.0%	21.0%

Keystrokes (works in chain or AOS mode): Firm A: $300 - 100 = \times .75 = \rightarrow \mathbf{150}$. Firm B: $100 \div 5 = +/- + 300 = \times .75 = \rightarrow \mathbf{210}$; then $\div 1000 = \rightarrow \mathbf{0.21}$.

Reading it: identical economics, a 6-point ROE gap. Firm B's equity is also \$60M higher (\$80M asset $\times (1 - 0.25)$), which lowers its P/B for the same market value.

Example 4 — Adjusting a GAAP firm's book value for R&D

A US firm reports equity of \$1,500M and a market capitalization of \$6,000M. Its R&D spending over the last five years (oldest first) was \$80M, \$90M, \$100M, \$110M, and \$120M. Treat R&D as an asset amortized straight-line over 5 years. The unamortized share is 100% for the current year, 80% for last year, then 60%, 40%, and 20%.

- Research asset = $120 + 110(0.8) + 100(0.6) + 90(0.4) + 80(0.2) = \320M
- Adjusted equity = $1,500 + 320 = \$1,820\text{M}$ (ignoring tax effects)
- Reported $P/B = 6,000 \div 1,500 = 4.00$; adjusted $P/B = 6,000 \div 1,820 = 3.30$

Keystrokes (AOS mode): $120 + 110 \times .8 + 100 \times .6 + 90 \times .4 + 80 \times .2 = \rightarrow \mathbf{320}$.

5. Common Pitfalls

1. **"Intangible" does not mean "not real."** Real vs. financial is about function; tangible vs. intangible is about physical form.
 2. **Assuming GAAP and IFRS treat development costs alike.** Research is expensed under both; qualifying development costs can be capitalized only under IFRS.
 3. **Amortizing goodwill.** Public-company goodwill is tested for impairment, not amortized, under both frameworks. Only finite-lived identifiable intangibles are amortized.
 4. **Treating goodwill as a plug you choose.** It is the residual *after* every identifiable asset is measured at fair value.
 5. **Dismissing an impairment as "just non-cash."** The timing and size are informative about earlier guidance and earnings quality.
 6. **Comparing P/B across accounting regimes unadjusted.** Capitalized development costs raise book value and lower P/B.
 7. **Chain vs. AOS on the BA II Plus.** Mixed addition and multiplication gives wrong answers in chain mode.
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6. Key Terms Checklist

☐ real asset ☐ financial asset ☐ intangible asset ☐ identifiable ☐ separable ☐ contractual-legal criterion ☐ goodwill ☐ finite-lived intangible ☐ indefinite-lived intangible ☐ amortization ☐ impairment ☐ purchase price allocation ☐ internally generated intangible ☐ capitalized development cost ☐ tangible book value ☐ price-to-book (P/B) ☐ quality of earnings ☐ comparability

7. Practice Questions (CFA-style, A-C)

Q1 (Conceptual). A firm's most valuable resource is a process patented after years of in-house research. Under both US GAAP and IFRS, the most likely balance-sheet treatment is that the patent is: - A. Recorded at fair value as an intangible asset. - B. Largely absent, because the costs were expensed as incurred (IFRS may capitalize qualifying development costs). - C. Recorded as a financial asset because it represents a legal claim.

Q2 (Algorithmic). A company reports equity of \$5,000M, goodwill of \$1,200M, other intangibles of \$300M, and 200M shares outstanding. The share price is \$28. Its price-to-tangible-book ratio is closest to: - A. 1.12 - B. 1.47 - C. 1.60

Q3 (Algorithmic). An acquirer pays \$600M for a target whose net tangible assets have a fair value of \$380M. Identifiable intangibles are valued at \$90M with a 9-year life. Annual intangible amortization and recorded goodwill are: - A. \$24.4M per year; goodwill \$130M - B. \$10.0M per year; goodwill \$130M - C. \$10.0M per year; goodwill \$220M

Q4 (Algorithmic). Two identical firms start the year with equity of \$800M and earn \$240M before development costs. Each spends \$60M on development. The GAAP firm expenses it; the IFRS firm capitalizes it and amortizes straight-line over 5 years. Tax rate is 25%. The IFRS firm's first-year ROE (on beginning equity) exceeds the GAAP firm's by about: - A. 0.0 percentage points - B. 4.5 percentage points - C. 6.0 percentage points

Q5 (Conceptual). A company books a large goodwill impairment two quarters after management said the acquired unit was "ahead of plan." An analyst assessing quality of earnings should view this primarily as: - A. Irrelevant, because the charge is non-cash. - B. A signal to investigate whether earlier guidance was poor or recognition was delayed. - C. Proof of accounting fraud.

Q6 (Conceptual). An IFRS reporter capitalizes development costs; a US peer expensing the same type of spending has the same market value. Compared with a like-for-like basis, the IFRS firm's unadjusted P/B will tend to look: - A. Higher - B. Lower - C. The same

Q7 (Algorithmic). A US firm has equity of \$1,000M and a market value of \$5,000M. Its R&D over the last five years (oldest first) was \$50M, \$60M, \$70M, \$80M, and \$90M. Capitalizing R&D and amortizing straight-line over 5 years (unamortized share 100%, 80%, 60%, 40%, 20% from newest to oldest), the adjusted P/B is closest to: - A. 3.70 - B. 4.07 - C. 5.00

8. Answer Key

Q	Answer	Explanation
1	B	Internally generated intangibles are expensed as incurred, so the asset never reaches the balance sheet. Acquired patents would be capitalized.
2	C	TBV $= 5,000 - 1,200 - 300 = 3,500$; per share 17.50; $28 \div 17.50 = 1.60$. A uses book value (1.12); B subtracts goodwill only (1.47).
3	B	Goodwill $= 600 - 380 - 90 = 130$; amortization $= 90 \div 9 = 10$. A amortizes goodwill too ($220 \div 9 = 24.4$); C lumps intangibles into goodwill.
4	B	GAAP: $(240 - 60)(0.75) = 135$, ROE 16.875%. IFRS: $(240 - 12)(0.75) = 171$, ROE 21.375%. Gap = 4.5 points. C forgets the tax effect ($48 \div 800 = 6.0$).
5	B	Impairments are non-cash but their timing is informative; this is a quality-of-earnings flag, not proof of fraud.
6	B	Capitalizing raises book equity, the P/B denominator, so P/B looks lower (the firm looks cheaper) than on a like-for-like basis.
7	B	Research asset $= 90 + 80(0.8) + 70(0.6) + 60(0.4) + 50(0.2) = 230$; adjusted equity 1,230; $5,000 \div 1,230 = 4.07$. A adds all R&D with no amortization (3.70).

9. Sources

Standards - IFRS Foundation, IAS 38 *Intangible Assets*: <https://www.ifrs.org/issued-standards/list-of-standards/ias-38-intangible-assets/> - FASB Codification, ASC 350 and ASC 805-20: <https://asc.fasb.org> (free registration) - FASB Invitation to Comment, *Agenda Consultation — Recognition of Intangibles*: <https://storage.fasb.org/ITC%E2%80%9494Recognition%20of%20Intangibles.pdf>

Big 4 guides - Deloitte, *A Roadmap to Goodwill and Intangible Assets Accounting*: <https://dart.deloitte.com/US/en/home/publications/deloitte-guidance/roadmap-ifrs-us-gaap-comparison/chapter-1-assets/1-5-intangible-assets> - Deloitte, *IFRS vs. US GAAP Comparison*, Ch. 1.5: <https://dart.deloitte.com/USDART/home/publications/deloitte-guidance/roadmap-ifrs-us-gaap-comparison/chapter-1-assets/1-5-intangible-assets> - KPMG, *Handbook: Impairment of Nonfinancial Assets*: <https://kpmg.com/us/en/frv/reference-library/2025/handbook-impairment-nonfinancial-assets.html> - PwC Viewpoint (free account): <https://viewpoint.pwc.com>

Analysis and textbooks - CFA Institute, *Investor Perspectives: Intangible Assets* (2025): https://rpc.cfainstitute.org/sites/default/files/docs/surveys/intangibles-report_online.pdf - Bodie, Kane & Marcus, *Investments* (13e), Ch. 1 (§1.1, real vs. financial assets) and Ch. 19 (comparability problems) - Brealey, Myers & Allen, *Principles of Corporate Finance* (real vs. financial assets) - Palepu & Healy, *Business Analysis and Valuation*

All numeric answers were checked with Python when drafted.